

Coordinate Descent with Randomized Coordinate Selection

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1 Algorithm Description

The algorithm iteratively updates the full parameter vector by computing the gradient along a randomly selected coordinate direction at each step. It uses this coordinate-wise gradient information to determine the update direction and step size for the entire parameter vector. It is designed for functions satisfying the Polyak-Łojasiewicz condition that are smooth with respect to individual coordinates.

1.1 Assumptions

The algorithm assumes:

- Polyak-Łojasiewicz (PL) condition: The function satisfies the PL condition, ensuring that the gradient provides sufficient information about the distance to the optimal function value.
- Coordinate-wise smoothness: The function is smooth with respect to individual coordinates, with bounded coordinate-wise Lipschitz constants.

2 Pseudocode and Dry Run

2.1 Mathematical Formulation

Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that is smooth with respect to each coordinate and satisfies the Polyak-Łojasiewicz (PL) condition. The goal is to minimize $f(\mathbf{w})$ where $\mathbf{w} \in \mathbb{R}^n$.

2.1.1 Update Rules and Equations

1. **Gradient Computation:** At each iteration k , choose a random index $i_k \in \{1, 2, \dots, n\}$. Compute the partial derivative (gradient) of f with

respect to the i_k -th coordinate:

$$g_{i_k} = \frac{\partial f}{\partial w_{i_k}}(\mathbf{w}^{(k)})$$

2. **Parameter Update:** Update the i_k -th component of the parameter vector:

$$w_{i_k}^{(k+1)} = w_{i_k}^{(k)} - \eta g_{i_k}$$

All other components remain unchanged:

$$w_j^{(k+1)} = w_j^{(k)}, \quad \forall j \neq i_k$$

3. **Hyperparameters:**

- Learning rate (η): Determines the step size for the update.
- Iteration count (K): Total number of iterations to execute.

2.2 Pseudocode

Algorithm CoordinateDescentWithRandomizedSelection

Input: Initial parameter vector $\mathbf{w}^0$, learning rate η , maximum iterations K

Output: Optimized parameter vector $\mathbf{w}^K$

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1: Initialize  $\mathbf{w} = \mathbf{w}^0$ 
2: for  $k = 0$  to  $K-1$  do
3:   Randomly select  $i_k$  from  $\{1, 2, \dots, n\}$ 
4:   Compute gradient  $g_{i_k} = \frac{\partial f}{\partial w_{i_k}}(\mathbf{w})$ 
5:   Update  $w_{i_k} = w_{i_k} - \eta g_{i_k}$ 
6: end for
7: return  $\mathbf{w}$ 
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2.3 Dry Run Example

Consider the quadratic function $f(w) = (w - 3)^2$.

2.3.1 Initialization

- Initial parameter: $w^0 = 0$
- Step size: $\eta = 0.1$
- Minibatch size: 1 (deterministic selection)

2.3.2 Iterations

Iteration 1:

- Gradient computation:

$$g(w^0) = \left. \frac{d}{dw}(w-3)^2 \right|_{w=0} = 2(w-3) \Big|_{w=0} = -6$$

- Parameter update:

$$w^1 = w^0 - \eta \cdot g(w^0) = 0 - 0.1 \times (-6) = 0.6$$

- Objective value:

$$f(w^1) = (0.6 - 3)^2 = 5.76$$

Iteration 2:

- Gradient computation:

$$g(w^1) = \left. \frac{d}{dw}(w-3)^2 \right|_{w=0.6} = 2(0.6 - 3) = -4.8$$

- Parameter update:

$$w^2 = w^1 - \eta \cdot g(w^1) = 0.6 - 0.1 \times (-4.8) = 1.08$$

- Objective value:

$$f(w^2) = (1.08 - 3)^2 = 3.6864$$

Iteration 3:

- Gradient computation:

$$g(w^2) = \left. \frac{d}{dw}(w-3)^2 \right|_{w=1.08} = 2(1.08 - 3) = -3.84$$

- Parameter update:

$$w^3 = w^2 - \eta \cdot g(w^2) = 1.08 - 0.1 \times (-3.84) = 1.464$$

- Objective value:

$$f(w^3) = (1.464 - 3)^2 = 2.359296$$

The algorithm iteratively approaches the minimum of the function $f(w)$ through successive updates along randomly chosen coordinate directions, demonstrating convergence towards the optimal solution.

3 Convergence Theory

3.1 Assumptions

- **Polyak-Łojasiewicz (PL) Condition:** A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the PL condition if there exists a constant $\mu > 0$ such that for all $x \in \mathbb{R}^n$, we have:

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*),$$

where f^* is the global minimum value of f .

- **Coordinate-wise Smoothness:** The function f is smooth with respect to individual coordinates, i.e., for each coordinate i , there exists a constant $L_i > 0$ such that for all $x \in \mathbb{R}^n$ and any perturbation h along the i -th coordinate, we have:

$$|f(x + he_i) - f(x) - \langle \nabla_i f(x), h \rangle| \leq \frac{L_i}{2} h^2,$$

where e_i is the i -th standard basis vector and $\nabla_i f(x)$ is the partial derivative of f with respect to the i -th coordinate.

- **Parameter Constraints:** The step size $\eta > 0$ is chosen such that it satisfies the stability condition with respect to the coordinate-wise Lipschitz constants L_i .

3.2 Main Convergence Theorem

Consider the iterative update $x^{k+1} = x^k - \eta \nabla_{i_k} f(x^k) e_{i_k}$, where i_k is a randomly selected coordinate at each iteration. Under the PL and coordinate-wise smoothness conditions, the algorithm converges to the global minimum x^* with the following properties:

1. **Rate of Convergence:** The expected function value satisfies:

$$\mathbb{E}[f(x^k) - f^*] \leq \left(1 - \frac{\eta\mu}{n}\right)^k (f(x^0) - f^*).$$

2. **Expected Reduction:** At each iteration, the expected reduction in function value is given by:

$$\mathbb{E}[f(x^{k+1})] \leq f(x^k) - \frac{\eta\mu}{n} (f(x^k) - f^*).$$

3. **Special Case:** If the function is quadratic and isotropic (i.e., $L_i = L$ for all i), the convergence rate is linear with a factor of $\left(1 - \frac{\eta\mu}{n}\right)$.

3.3 Theoretical Properties

- **Stability Analysis:** The update is stable if the step size η satisfies $\eta < \frac{2}{L_i}$ for all i . This ensures that the algorithm does not diverge due to excessively large updates.
- **Sensitivity to Hyperparameters:** The convergence rate is sensitive to the choice of η . A smaller η may lead to slower convergence, while a larger η risks instability.
- **Comparison to Baseline Methods:** Compared to full gradient descent, coordinate-wise gradient descent can be more efficient in practice due to reduced per-iteration computational cost, especially when n is large.

3.4 Discussion

The developed convergence theory reveals that the proposed algorithm is effective under the Polyak-Łojasiewicz condition, achieving a linear convergence rate. The coordinate-wise smooth assumption ensures that updates are well-behaved with respect to individual parameters, which is particularly beneficial in high-dimensional settings. The sensitivity to the step size highlights the importance of parameter tuning to balance convergence speed and stability. Future work could explore adaptive step size strategies to further enhance the algorithm's robustness and performance.

4 Convergence Proof

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

1. **Polyak-Łojasiewicz (PL) Condition:** A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the PL condition if there exists a constant $\mu > 0$ such that for all $x \in \mathbb{R}^n$:

$$\frac{1}{2} \|\nabla f(x)\|^2 \geq \mu(f(x) - f^*),$$

where f^* is the global minimum value of f .

2. **Coordinate-wise Smoothness:** For each coordinate i , there exists a constant $L_i > 0$ such that for all $x \in \mathbb{R}^n$ and any perturbation h along the i -th coordinate:

$$|f(x + he_i) - f(x) - \langle \nabla_i f(x), h \rangle| \leq \frac{L_i}{2} h^2,$$

where e_i is the i -th standard basis vector and $\nabla_i f(x)$ is the partial derivative of f with respect to the i -th coordinate.

4.1.2 Lemmas

1. **Gradient Descent Lemma:** For a function f that is L -smooth, the following holds:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

4.2 Main Proof (Step-by-Step Derivation)

Consider the iterative update:

$$x^{k+1} = x^k - \eta \nabla_{i_k} f(x^k) e_{i_k},$$

where i_k is a randomly selected coordinate at each iteration.

4.2.1 Step 1: Coordinate-wise Update Analysis

For the selected coordinate i_k , the update can be expressed as:

$$x^{k+1} = x^k - \eta \nabla_{i_k} f(x^k) e_{i_k}.$$

The function value at x^{k+1} is:

$$f(x^{k+1}) \leq f(x^k) + \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \frac{L_{i_k}}{2} \|x^{k+1} - x^k\|^2.$$

Substitute $x^{k+1} - x^k = -\eta \nabla_{i_k} f(x^k) e_{i_k}$:

$$f(x^{k+1}) \leq f(x^k) - \eta \langle \nabla f(x^k), \nabla_{i_k} f(x^k) e_{i_k} \rangle + \frac{L_{i_k}}{2} \eta^2 (\nabla_{i_k} f(x^k))^2.$$

4.2.2 Step 2: Expectation over Random Coordinate Selection

Taking expectation over the random choice of i_k :

$$\mathbb{E}[f(x^{k+1})] \leq f(x^k) - \frac{\eta}{n} \|\nabla f(x^k)\|^2 + \frac{\eta^2}{2n} \sum_i = 1^n L_i (\nabla_i f(x^k))^2.$$

4.2.3 Step 3: Applying the PL Condition

Using the PL condition:

$$\frac{1}{2} \|\nabla f(x^k)\|^2 \geq \mu(f(x^k) - f^*).$$

Thus:

$$\mathbb{E}[f(x^{k+1})] \leq f(x^k) - \frac{2\eta\mu}{n} (f(x^k) - f^*) + \frac{\eta^2}{2n} \sum_i = 1^n L_i (\nabla_i f(x^k))^2.$$

4.2.4 Step 4: Stability Condition

For stability, choose η such that:

$$\eta < \frac{2}{L_i} \quad \text{for all } i.$$

4.3 Rate Derivation (With Constants)

Assuming η is small enough, the term involving η^2 becomes negligible:

$$\mathbb{E}[f(x^{k+1}) - f^*] \leq \left(1 - \frac{2\eta\mu}{n}\right) (f(x^k) - f^*).$$

4.4 Special Cases

1. **Quadratic and Isotropic Functions:** If f is quadratic and $L_i = L$ for all i , the convergence rate simplifies to:

$$\mathbb{E}[f(x^k) - f^*] \leq \left(1 - \frac{\eta\mu}{n}\right)^k (f(x^0) - f^*).$$